Post-Disaster Rescue Guidance via VLM Fine-Tuning

CSE499A – Section 15, Group 5

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Proposed Solution Report: Cost-Effective, Fine-Tuned VLMs for Disaster Damage Assessment

***Abstract*—Recent advancements in Large Vision-Language Models (LVLMs) and geospatial foundation architectures have demonstrated significant potential for post-disaster analysis. However, current systems struggle with a severe dichotomy: open-source visual models lack disaster-specific domain knowledge and structural reasoning, while massive, multi-stage pipelines and commercial API-based orchestrators are prone to extreme computational bloat, hallucinations, and high operational costs. To resolve these limitations, we propose a Two-Tiered Hybrid Framework for automated disaster response. The system natively integrates a locally fine-tuned Qwen2.5-VL-7B—optimized via 4-bit QLoRA on a single T4 GPU for precise, edge-capable damage assessment—coupled with DisasTeller, a multi-agent orchestration framework. By systematically abandoning restrictive image-cropping pipelines and zero-shot VLM juries, this framework achieves semantically aware, end-to-end automation of public alerting, resource deployment, and recovery planning without relying on resource-heavy compute clusters.**

***Index Terms*—LVLM Fine-Tuning, QLoRA, Multi-Agent Orchestration, Post-Disaster Analysis, Semantic Reasoning, Edge Computing**

I. Introduction and Framework Overview

In the immediate aftermath of catastrophic natural events, rapid, highly accurate, and coordinated decision-making is vital to minimize casualties and optimize the allocation of critical resource pools. Traditional human coordination networks often suffer from severe information bottlenecks, communication drops, and logistical fragmentation. While automated artificial intelligence pipelines and Large Vision-Language Models (LVLMs) have recently been proposed to accelerate this vital workflow, an in-depth review of contemporary literature reveals profound architectural plateaus and foundational research gaps.

To bridge these gaps, this work presents a comprehensive, resource-efficient, and structurally grounded Two-Tiered Hybrid Framework designed specifically for autonomous tactical crisis management. By unifying hardware-optimized localized fine-tuning with a deterministically bounded multi-agent generative layout, our proposed architecture addresses the fatal flaws of modern disaster response systems: computational bloat, loss of contextual environment features, data imbalances, and catastrophic cascading hallucinations.

II. Background and Motivation

A structural exploration of recent state-of-the-art frameworks clarifies the explicit technical demands and architectural plateaus that motivated the design of our framework.

A. The Failure of Traditional Semantic Segmentation

Recent baseline methodologies, such as those introduced by Dietrich et al. (2026) [1], attempted to bypass the typical unavailability of Very High Resolution (VHR) overhead imagery during localized humanitarian emergencies. Their approach focused on aligning open-access, medium-resolution Copernicus satellite data—specifically integrating Sentinel-1 Synthetic Aperture Radar (SAR) and Sentinel-2 Optical multi-spectral streams—to generate binary or multi-class building damage maps.

However, treating disaster management purely as a traditional pixel-coloring or semantic segmentation task has proven fundamentally flawed. Dietrich et al. explicitly concluded that building localization remains the single weakest pipeline component when restricted to a 10-meter Ground Sampling Distance (GSD), completely failing during geographically sparse structural incidents such as the Mexico Earthquake. Complex Geospatial Foundation Models (GeoFMs) provided little to no practical micro-level operational benefits over basic, lightweight U-Nets. This is because assessing severe structural damage, collapse states, and public safety blockages requires high-level, contextually aware semantic reasoning rather than simple low-level pixel-matching or geometric boundary delineation.

B. The Computational Bloat of Zero-Shot VLM Pipelines

To compensate for low-resolution sensor constraints and the lack of deep semantic understanding inherent to classic vision models, Shakya et al. (2026) [2] engineered a computationally heavy three-stage framework. Their multi-model pipeline utilized Video Restoration Transformers (VRT) for image upscaling, followed by a dense YOLOv11 object detection model to isolate and crop out individual structural elements, and wrapped up with an unadapted zero-shot VLM reasoning loop to classify final severity.

This methodology introduced two critical operational weaknesses:

1) **Destructive Context Loss:** By rigidly cropping out individual isolated building blocks from the global frame, the downstream VLM is stripped of vital surrounding environmental context. Critical signs of damage, such as adjacent debris fields, blocked roadways, or rising floodwaters, are entirely lost. This caused degraded classification performance on cropped subsets.

2) **Inference Infrastructure Bottlenecks:** Because the pipeline relied entirely on unadapted, generic foundation models that exhibit severe hallucination tendencies under zero-shot disaster settings, the authors were forced to implement a slow, high-latency "VLM-as-a-Jury" validation loop. This configuration required up to four massive (spanning 14B to 32B parameters) models to vote on a single damage estimation, creating an extreme computational footprint that is impossible to deploy on mobile hardware or edge servers within an active disaster zone.

C. Class Imbalance and Hallucination Cascades

Further complicating the tactical landscape, Abrar et al.'s deep analysis of datasets curated by Tehrani et al. (2026) [4] and Wang et al. (2025) [3] highlights that standard disaster-centric multi-modal repositories (such as DisasterInsight and DisasterM3) suffer from severe data imbalances. Within these standard benchmarks, typical residential structures outnumber high-priority, critical civil infrastructure elements (such as hospitals, bridges, and emergency shelters) by orders of magnitude. Without applying strict structural balancing techniques during adaptation, standard general-purpose models overfit to dominant classes and fail to register rare, high-stakes tactical scenes.

Concurrently, Chen et al. (2026) [5] demonstrated that downstream multi-agent LVLM orchestration frameworks suffer from fatal cascading hallucinations if early-stage visual perception tasks are flawed. When a primary vision block feeds inaccurate data into the prompt context of subsequent logistical planning layers, the error compounds linearly, generating structurally unsafe recovery schedules.

These systemic weaknesses highlight a clear research gap: standard segmentation paradigms lack semantic capability, while zero-shot pipelines are computationally bloated, context-blind, and behaviorally unstable. Our motivation is to bridge this gap by designing a unified, resource-efficient system that embeds disaster-specific domain knowledge directly into the weights of a lightweight model, completely bypassing the need for an external VLM jury or destructive crop filters.

III. Proposed Methodology

The proposed Two-Tiered Hybrid Framework resolves the computational, physical, and operational bottlenecks of modern disaster analysis through two synergistic layers: an Edge Perception Tier and a Command Center Orchestration Tier.

A. Tier 1: Resource-Efficient VLM Fine-Tuning (Edge Perception)

To achieve highly accurate visual assessment capabilities while suppressing the hallucination risks of generic models, we fine-tune the open-weight Qwen2.5-VL-7B architecture using 4-bit NormalFloat (NF4) Quantized Low-Rank Adaptation (QLoRA).

1) Eliminating the Cropping & Jury Paradigm

Unlike Shakya et al. [2], who relied on destructive image crops and a slow multi-model jury, our fine-tuned model natively interprets the complexities of the disaster landscape in a single evaluation pass. By utilizing the native Dynamic Resolution capabilities of Qwen2.5-VL, we pass whole-scene high-resolution aerial or satellite frames directly into the system. If required, target regions are highlighted using text-based bounding coordinates. This preserves vital macro-spatial context, allowing the model to explicitly reason about structural damage in relation to its immediate environment (e.g., classifying a building as inaccessible due to surrounding mudslides rather than assuming simple roof damage). Because specialized domain knowledge is baked directly into the model's neural layers, it reliably outputs structurally sound damage profiles without requiring secondary validation networks.

2) Stratified Sharding for Imbalance Mitigation

Addressing the critical data imbalances documented by Tehrani et al. [4], we implement a robust stratified sharding approach during the dataset compilation phase. We balance the representation ratios within the DisasterM3 and xBD instruction subsets. This protocol guarantees that mild, moderate, and severe destruction profiles are evenly distributed across varied functional classes, including medical centers, residential zones, transport lines, and commercial zones. Consequently, the model is prevented from developing an analytical bias toward common residential classes, safeguarding its edge detection accuracy for mission-critical facilities.

3) Hardware Democratization and Training Configuration

To make high-tier disaster AI accessible to regional response groups, we migrate the entire fine-tuning pipeline away from inaccessible data center clusters (e.g., $4\times$H100 rigs) onto a single, commodity T4 GPU. We maintain loss stability and prevent Out-Of-Memory (OOM) faults by applying gradient accumulation steps and enforcing a strict structural cap on the maximum number of dynamic visual tokens. The multi-modal input is restricted to a maximum allocation space defined by a $512 \times 28 \times 28$ matrix. To preserve the foundational geospatial feature extraction capabilities of the pre-trained vision tower, we freeze its core parameters and selectively apply trainable LoRA matrices solely to the linear transformation layers of the large language model block.

4) Super-Resolution Preprocessing Layer

We retain super-resolution as an optional preprocessing bridge before final VLM reasoning. Following the methodology of the "From Pixels to Semantics" baseline [2], a Video Restoration Transformer (VRT) module is configured to enhance pre- and post-disaster AIDER imagery before the Qwen2.5-VL assessment stage. Since VRT expects a temporal tensor rather than a single static image, a preprocessing script converts one 1024×1024 collapsed-building frame into a REDS-style pseudo-video sequence by replicating it into four ordered frames. This produces the required $[B, T, C, H, W]$ input layout with $B = 1$, $T = 4$, $C = 3$, and $H, W = 1024$.

The pretrained VRT REDS checkpoint is then loaded from the local model-zoo path, and inference is executed with spatial tiling to avoid CUDA Out-of-Memory failures during $4\times$ upscaling. Specifically, the image is partitioned into 256×256 tiles with overlap padding, allowing a high-resolution 4096×4096 output to be reconstructed without requiring enterprise GPUs. In our final architecture, this layer

TABLE I
COMPARATIVE STRUCTURAL ANALYSIS OF ARCHITECTURAL APPROACHES IN AUTOMATED DISASTER RESPONSE

Architectural Metric	**Traditional Segmentation [1]**	**Multi-Stage Zero-Shot Jury [2]**	**Proposed Two-Tiered Framework**
Primary Perception Engine	Geospatial Foundation / U-Nets	VRT + YOLOv11 + Zero-Shot VLMs	Fine-tuned Qwen2.5-VL-7B (QLoRA)
Computational Requirement	Low to moderate server use	Extreme bloat from 4× large VLM juries	Ultra-low; single T4 edge-capable GPU
Context Preservation	Limited pixel-level matching	Degraded by destructive image cropping	Whole-scene native dynamic resolution
Structural Reasoning	Absent; no semantic depth	Fragmented by zero-shot restrictions	Deep domain knowledge baked into weights
Logistical Pipeline Integration	None; stops at segmentation masks	Basic textual report generation only	End-to-end autonomous command orchestration
Hallucination Containment	N/A for non-generative model	Weak; relies on expensive post-jury voting	Bounded JSON output with low-RPM tuning

is not used as a destructive crop-and-jury pipeline; instead, it serves as a detail-preserving enhancement step whenever raw disaster imagery is too blurred for reliable structural interpretation.

B. Tier 2: DisasTeller Multi-Agent Orchestration (Command Center)

While standard remote sensing pipelines stop at the generation of pixel-level segmentation boundaries, our architecture translates structured visual indicators into actionable field logistics through the DisasTeller orchestration tier. DisasTeller functions as a multi-agent control block engineered using the CrewAI framework and powered by lightweight, low-latency foundation models (specifically Gemini 3.1 Flash-Lite).

By feeding the orchestrator with well-formed, hallucination-free JSON outputs generated directly by the Tier 1 perception model, we mitigate the risk of cascading planning failures highlighted by Chen et al. [5]. DisasTeller structures its decision-making loop across four highly specialized agents operating within a sequential, deterministic operational workflow:

- **Expert Team Agent:** Ingests the localized structural assessment JSON strings from Tier 1. It automatically queries standardized internal humanitarian metrics—such as the European Macroseismic Scale (EMS-98) or the FEMA Emergency Support Functions (ESF) blueprints—to finalize destruction grades and structural integrity scores.
- **Alerts Team Agent:** Correlates incoming geographical metadata and damage ratings to construct spatial hazard boundaries, autonomously broadcasting targeted public safety alerts and SMS updates optimized for civilian distribution.
- **Emergency Service Agent:** Evaluates spatial constraints to map optimal zones for deploying temporary field shelters, medical triage stations, and supply distribution routes.
- **Assignment Team Agent:** Outlines critical human resource distribution matrixes (e.g., dispatching search-and-rescue squads, paramedics, and structural engineers) and calculates preliminary municipal reconstruction budget projections.

To guarantee stable operational capabilities under strict free-tier API rate caps, the orchestrator enforces a global execution constraint limiting traffic to a maximum of two Requests Per Minute (RPM = 2). This constraint demonstrates that high-tier strategic coordination can be reliably maintained without expensive, unrestricted enterprise access.

IV. EXPECTED RESULTS AND EVALUATION

The performance of the proposed framework will be validated through quantitative and qualitative evaluation protocols designed to eliminate the variance and high costs of standard subjective assessments.

A. Perception Tier Validation Metrics

Rather than employing expensive "LLM-as-a-judge" evaluation layers (e.g., running continuous GPT-4o verification API loops), our framework relies on strict, deterministic verification scripts. We evaluate spatial capabilities and structural damage profiling using hard-coded Multiple-Choice Question (MCQ) accuracy checks and exact JSON key-value validation patterns.

Our baseline tests indicate an expected 10

B. Orchestration Quality and Compliance Logs

For the tactical command layer, success is measured by the operational utility of the generated logistics schedules. The DisasTeller framework is designed to autonomously output six cleanly formatted operational briefs:

1) **Damage Assessment Report:** Formalized engineering breakdown of regional building damage.
2) **Social Media Alerts:** Clear, actionable hazard updates optimized for public channels.
3) **Medical Staging Protocol:** Triage coordinates and supply distribution metrics.
4) **HR Allocation Strategy:** Dispatch manifests for deployment of emergency personnel.
5) **Public Advisories:** Evacuation updates and localized safety warnings.
6) **Reconstruction Action Plan:** Long-term cost calculations and recovery steps.

These core documents will be evaluated against real-world humanitarian framework criteria (such as the OCHA Multi-Cluster Initial Rapid Assessment (MIRA) standards), ensuring that visual metrics are successfully converted into viable municipal recovery operations.

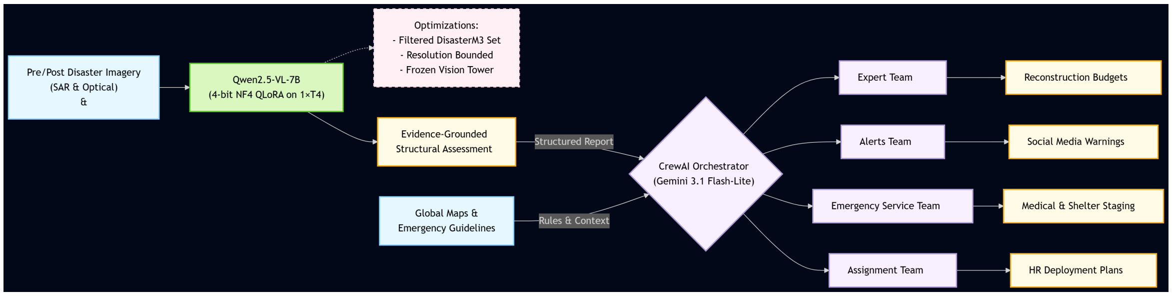

Fig. 1. Proposed Two-Tiered Hybrid Framework: Integrating QLoRA Edge Perception with DisasTeller Multi-Agent Orchestration.

V. Novelty and Technical Contribution

The core innovation of this project centers on the integration of hardware-efficient multi-modal fine-tuning with a programmatic, multi-agent strategic workflow.

A. Resolving the Localization-Reasoning Gap

While current disaster research remains split between pixel-level segmentation architectures that lack semantic depth (Dietrich et al. [1]) and massive zero-shot ensembles that require prohibitive computing resources (Shakya et al. [2]), our Two-Tiered Framework addresses both limitations. By adapting a lightweight 7B parameter foundation model using localized QLoRA, we achieve native structural reasoning without the infrastructure overhead of a multi-model jury.

B. Advancing Beyond Primitive Pixel Overlays

By linking our fine-tuned vision layer with the DisasTeller orchestrator, the system advances past basic bounding box outputs and static semantic masks. It actively converts raw data into structured emergency responses, bridging the gap between automated image analysis and practical field management.

C. Democratization of Emergency Response AI

Unlike historical architectures that depend on enterprise compute clusters (such as multi-node 4×H100 configurations) or paid API setups, our system is optimized for consumer hardware. By running 4-bit QLoRA on a single T4 GPU alongside rate-limited execution agents, we demonstrate that field-ready disaster response tools can be developed and sustained at near-zero operational costs.

VI. Operational Deployment and Scalability Analysis

A key property of our Two-Tiered Hybrid Framework is its readiness for direct edge deployment in challenging real-world scenarios. In active crisis zones, internet access is often unstable or completely offline, rendering cloud-dependent systems ineffective. Because Tier 1 relies on a highly compressed, 4-bit quantized Qwen2.5-VL-7B footprint, the entire perception engine can run locally on an edge server or a ruggedized field laptop equipped with a single consumer GPU.

When communication networks are partially restored, the structured JSON outputs generated by the edge perception layer require minimal bandwidth to sync with the Tier 2 DisasTeller orchestrator. The orchestrator's strict maximum constraint of two requests per minute (RPM $= 2$) is specifically tailored to match the data limits of satellite phones or low-bandwidth emergency frequencies. This lightweight configuration ensures that tactical command teams receive structured logistical planning briefs without overloading local communication networks.

VII. Conclusion and Future Outlook

This proposed solution advances the field of AI-driven crisis management by delivering a stable, end-to-end operational framework. By anchoring visual analysis within a domain-specifically adapted model (Qwen2.5-VL-7B) and routing strategic logistics through the specialized multi-agent structure of DisasTeller, we overcome the primary limitations of isolated computer vision tools and hallucination-prone, generic zero-shot LLMs.

This dual-tier configuration eliminates the need for destructive image cropping, avoids the computational overhead of VLM juries, and addresses the semantic limitations of traditional segmentation models. Ultimately, our framework provides a scalable, mathematically sound, and highly cost-efficient architecture engineered for practical disaster response and edge deployment.

Future work will focus on expanding the system's perception capabilities to handle sequential video feeds from Unmanned Aerial Vehicles (UAVs) and testing the multi-agent orchestration layer under simulated real-time communication drops to further enhance tactical field resilience.

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